Monthly crime trends: a global STL-style decomposition

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What this project estimates

This is a descriptive decomposition of monthly crime counts for every city in the RTCI national sample. Rates are annualized from monthly counts: a monthly count `m` in a population `N` is shown as `12 * m / N * 100,000`. This puts homicide on the familiar annual rate scale rather than below one.

The global model is an aggregated-binomial smooth decomposition. For crime type `c`, city `i`, and month `t`, its global probability is:

`logit(p_global[c,t]) = alpha[c] + trend[c,t] + seasonal[c,month(t)]`.

The city fit then adds two empirical intercept terms to that global logit: `city_effect[i]`, the population-weighted mean departure across crimes and months, and `city_crime_effect[i,c]`, the population-weighted departure left after the city effect. Thus:

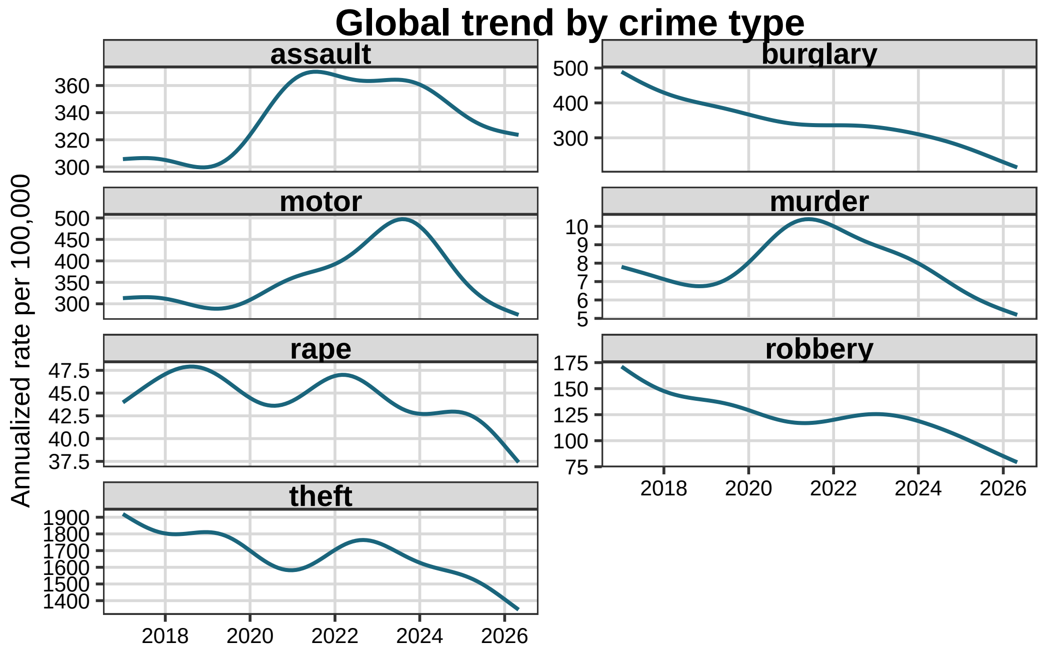
`logit(p_city[i,c,t]) = logit(p_global[c,t]) + city_effect[i] + city_crime_effect[i,c]`.

The city-by-crime-by-month overdispersion term is the observed cell departure from this fitted city baseline. It is centered at zero within every city and crime type and retained at the complete time resolution. It is not a Pearson residual. The global residual is centered at zero within crime type.

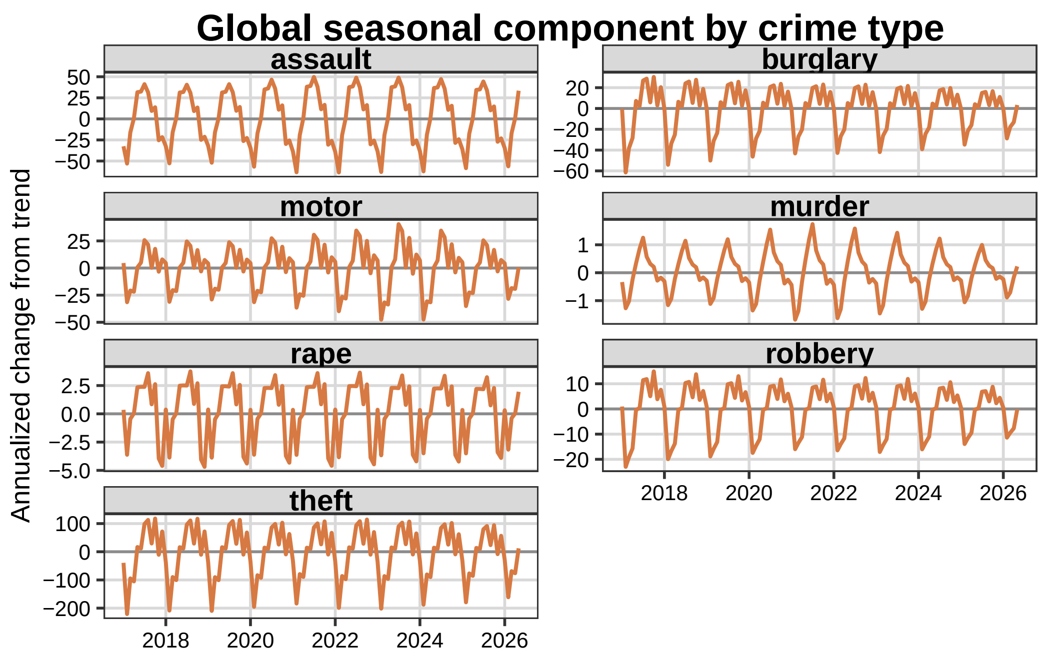
Data coverage

The deliverable contains 458,392 valid city-month-crime observations from 586 cities. There is no population threshold.

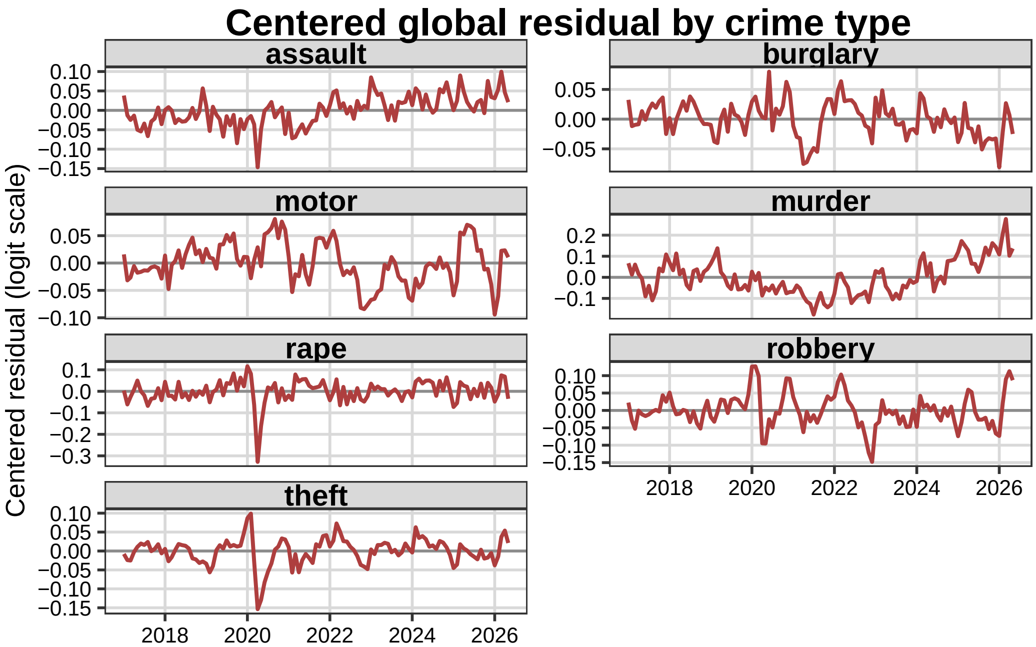
Global trend



Global seasonal component



Global residual



All three figures use one readable panel per crime type. The complete city-by-crime-by-month decomposition and the city-name crosswalk are available in the app data; the app has crime selection, city detail, and a Leaflet map.